DC Electronic Load Rev E: block diagram

Control board + power board on one 2-layer panel, isolated, USB-C powered. 150 W, 0.6 - 58 V, 0 - 10.1 A. Hardware CC and CV, firmware CR / CP / battery. OLED + encoder UI, USB CDC / DFU, SCPI. JLCPCB / LCSC parts.
Each block is a sheet (page number under its name); every line is a wire between sheet pins, colour-coded by net class (key bottom left). Power rails are global symbols on the sheets.
The analog stage is the Rev C design re-simulated with the TLV9154 (../simulation/rev_e); check_netlist_reve.py proves the netlist against deck 06.

USB SIDE (DGND, earth-referenced through the PC)

USB & Power

USB_DM
USB_DP
CC1_SENSE
CC2_SENSE
FAN_EN
FAN_PWM
FAN_TACH

File: sheets/usb_power.kicad_sch

MCU

USB_DM
USB_DP
CC1_SENSE
CC2_SENSE
FAN_EN
FAN_PWM
FAN_TACH
SPI1_SCK
SPI1_MOSI
SPI1_MISO
CS_DAC
CS_ADC
LOAD_OFF

File: sheets/mcu.kicad_sch

Board Link

SPI1_SCK
SPI1_MOSI
SPI1_MISO
CS_DAC
CS_ADC
LOAD_OFF
SPI1_SCK_LK
SPI1_MOSI_LK
SPI1_MISO_LK
CS_DAC_LK
CS_ADC_LK
LOAD_OFF_LK

File: sheets/board_link.kicad_sch

Front Panel UI

I2C_SCL
I2C_SDA
LED_ON
LED_FAULT
ENC_A
ENC_B
ENC_SW
BTN_ON
BTN_MODE
BTN_MENU

File: sheets/ui.kicad_sch

ISOLATION BARRIER, 4 mm

ISOLATED ANALOG SIDE (AGND = DUT minus, floats with the DUT)

Isolated Interface

SPI1_SCK_LK
SPI1_MOSI_LK
SPI1_MISO_LK
CS_DAC_LK
CS_ADC_LK
LOAD_OFF_LK
LOAD_OFF_ISO
TEMP_MON
VSET
LOAD_CTRL
CV_SET
ISEN_ADC
VSEN_ADC

File: sheets/iso_interface.kicad_sch

Protection

LOAD_OFF_ISO
TEMP_MON
KILL
DRAIN
OTP_GATE
OFF_GATE
VSEN_OA

File: sheets/protection.kicad_sch

Setpoint & CV

VSET_MON
LOAD_CTRL
CV_SET
OTP_GATE
OFF_GATE
VSEN_OA_OVP
VSET
VSEN_OA

File: sheets/setpoint_cv.kicad_sch

Sense

ISEN_ADC
VSEN_ADC
VSEN_OA
SENSE_P
SENSE_N
AVG

File: sheets/sense.kicad_sch

Power Path

SENSE_P
SENSE_N
DRAIN

File: sheets/power_path.kicad_sch

Channel 1 / Channel 2 / Channel 3 / Channel 4

KILL
VSET
AVG
DRAIN
KILL_T
VSET_T
AVG_T
DRAIN_T

File: sheets/channel.kicad_sch

NET COLOUR KEY

- DUT current path
- set-point (LOAD_CTRL, V_SET, CV_SET)
- measurement (AVG, ISEN, VSEN, V_NTC)
- protection (KILL, OTP, OFF, LOAD_OFF)
- isolated SPI
- USB data
- supply rails (on the sheets)
- isolation barrier

How to read this page: signals flow left to right. The 12-way ribbon (page 5) carries the SPI and the 5 V / 3.3 V rails from the control board to the power board; the isolator, DAC and ADC (page 6) straddle the barrier. V_SET (page 7) is the one node the four channels follow; KILL (page 9) stops them; the source-node average AVG returns to the sense amplifier (page 8); DRAIN comes from the terminals (page 10) and ends at the dropout comparator. The channels (pages 11-14, sheets/channel.kicad_sch) are drawn as a chain: a *_T pin is the same net leaving on the right. PGND / AGND / DGND and the rails are global power symbols on the sheets.

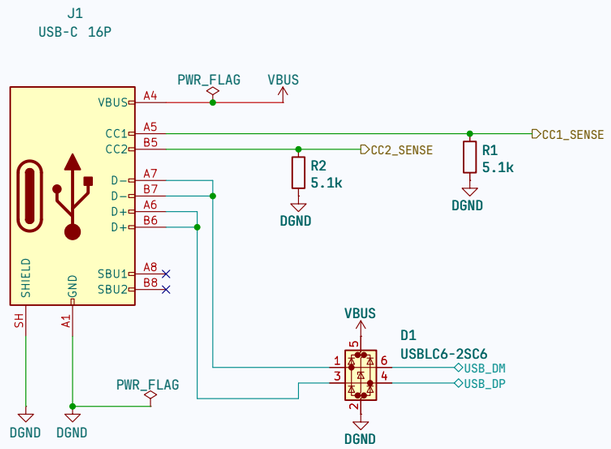
DC Load Box	
Sheet: /	
File: DC_Load_RevE.kicad_sch	
Title: DC Electronic Load Rev E: block diagram	
Size: A3	Date: 2026-11-03
KiCad E.D.A. 10.0.6	Id: 1/14

Rev: E

USB-C input and power tree

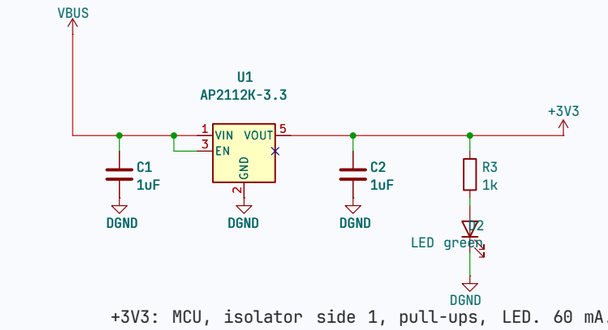
One USB-C port feeds three rails and the 5 V blower. Left of the barrier is USB ground (DGND); the analog stage floats on AGND = DUT minus.

A USB-C input: 5 V, data, current advertisement (CC)

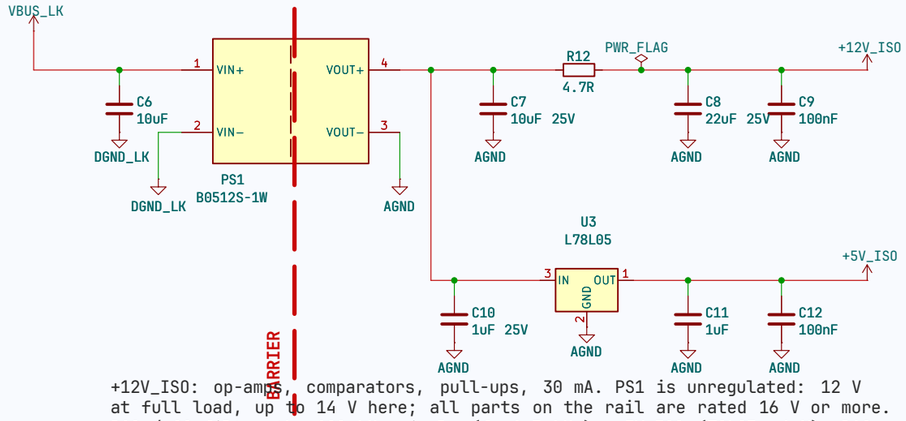


CC1 / CC2 5.1k pull-downs = USB-C sink. The MCU reads the CC voltage to learn the source limit: 0.4 V = 500 mA (legacy source), 0.9 V = 1.5 A, 1.7 V = 3 A. The whole box draws about 0.45 A with the blower at full speed. Shield to DGND: a metal enclosure is on the USB side; DUT terminals, heatsink hardware and the NTC lug must not touch it.

B Power tree: VBUS 5 V trunk, two branches, three rails

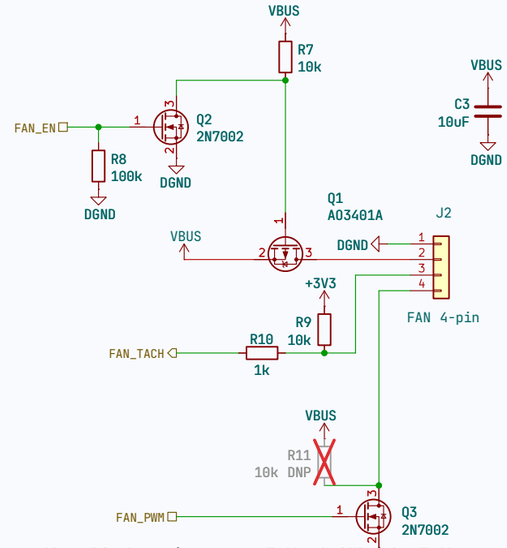


+3V3: MCU, isolator side 1, pull-ups, LED. 60 mA.



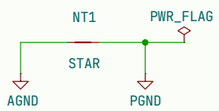
+12V_ISO: op-amps, comparators, pull-ups, 30 mA. PS1 is unregulated: 12 V at full load, up to 14 V here; all parts on the rail are rated 16 V or more. R12 / C8 filter the 100 kHz ripple (fc 1.5 kHz). +5V_ISO (78L05, 4 %): DAC, ADC, isolator side 2 and every threshold divider, so OTP, dropout, NTC-open and OVP thresholds do not depend on its absolute value; the DAC and ADC carry their own references. VBUS_LK / DGND_LK: VBUS / DGND after the J10-J11 cable. VERIFY the SIP-4 pin order of the module you buy.

C Blower switch and connector



J2 = PC fan pinout on 5 V: 1 GND, 2 +5 V switched by Q1, 3 tach (open collector, pulled to 3.3 V), 4 PWM 25 kHz. A 5 V 5015 blower (2 wires) uses pins 1 and 2 and is on/off through FAN_EN; a 4-wire fan gets PWM from Q3 (open-drain, firmware inverts the duty). R11 only for fans without a PWM pull-up. C3 is the blower bulk capacitor: keep the motor noise on its own VBUS trace.

D Ground star point and the isolation barrier rating



AGND (analog reference) meets PGND (DUT minus, source resistors) ONLY at NT1. Place NT1 at the centre of the four Rs bottom pads: 1 mR between an Rs and NT1 is 10 mV at 10 A (4 % of full scale).

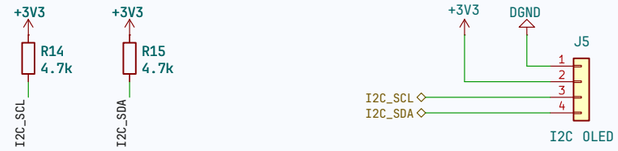
Barrier: DGND side to AGND / PGND side. Rated 150 V DC working (IEC 60664-1 functional, PD2, material group IIIb: 1.6 mm creepage; IPC-2221B B2: 0.7 mm). Board rule 4 mm, no copper crosses on any layer. CA-IS3761 (5 kVrms, 8 mm) and the 1.5 kVDC DC-DC exceed it by far. Float the load on a DUT referenced within 150 V of earth; not for mains work.

DC Load Box		
Sheet: /USB & Power/		
File: usb_power.kicad_sch		
Title: USB-C input and power tree		
Size: A3	Date: 2026-11-03	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 2/14

Front panel: OLED module, encoder, 3 buttons, 2 status LEDs

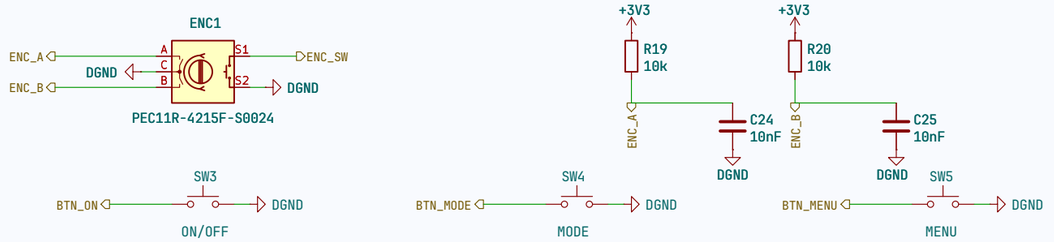
The readout is an off-the-shelf I2C OLED module on a 4-wire lead. Everything on this sheet is on DGND (USB side).

A OLED module header: 1.3" SH1106 or 0.96" SSD1306, I2C address 0x3C, 3.3 V



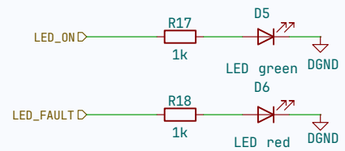
The module shows V, A, W, the set value, mode and status; menus for limits, fan and calibration. Short lead (under 15 cm) to the bezel. 1.3" SH1106 modules are the readable choice at arm's length; the 0.96" SSD1306 is the same pinout and protocol family. R14 / R15 pull up the bus here because the module may or may not carry its own.

B Encoder and buttons (MCU pull-ups on the buttons, RC on the encoder)



ON/OFF toggles the sink. MODE cycles CC/CV/CR/CP/BATT. MENU: limits, fan, cal. Push = digit select.

C Status LEDs: ON (green) while the load sinks, FAULT (red) for OTP / OVP / NTC-open / OFF



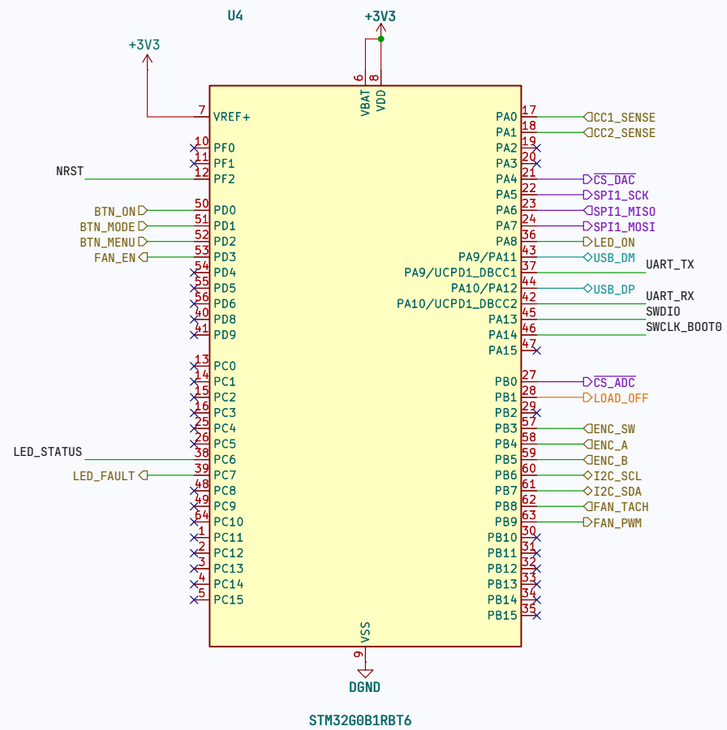
Panel LEDs beside the display: ON = the load is sinking, FAULT = a hardware trip or the firmware latch. The heartbeat LED is on the MCU sheet. The mode (CC/CV/CR/CP/BATT) and REMOTE are shown on the OLED.

DC Load Box		
Sheet: /Front Panel UI/		
File: ui.kicad_sch		
Title: Front panel: OLED module, encoder, 3 buttons, 2 status LEDs		
Size: A3	Date: 2026-11-03	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 3/14

MCU: STM32G0B1RBT6 (LQFP64, the G0B1 JLC stocks), USB device, SPI to the isolated stage, I2C to the OLED

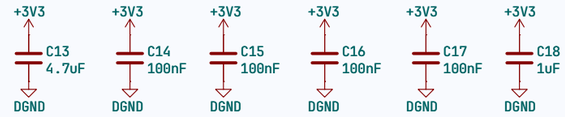
Crystal-less: USB SOF trims the 48 MHz HSI. DFU over USB is in ROM (hold BOOT, tap RESET). Everything here is on DGND.

A Microcontroller



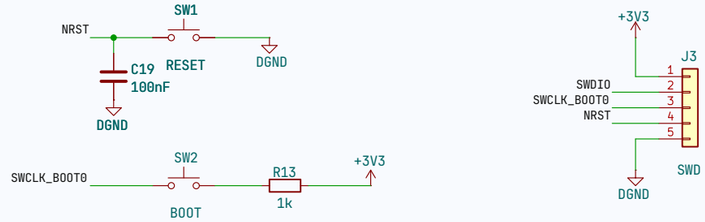
Pin plan (STM32G0B1RBT6, LQFP64; the CBT6 of Rev D is not stocked at JLC, same die, same pin names)
SPI1 PA5 SCK, PA6 MISO, PA7 MOSI, PA4 /CS DAC, PB0 /CS ADC, PB1 LOAD_OFF (isolator)
I2C1 PB6 SCL, PB7 SDA: the OLED module (page 4). No display driver, no level shifter.
TIM3 PB4 / PB5 encoder, PB3 push. TIM17 PB9 fan PWM, TIM16 PB8 tach, PD3 FAN_EN
ADC PA0 / PA1 = CC1 / CC2. USB PA11 DM, PA12 DP, crystal-less (CRS)
USART1 PA9 / PA10 debug. PD0 ON/OFF, PD1 MODE, PD2 MENU. PC6 heartbeat LED, PA8 ON LED, PC7 FAULT LED
Why not the STM32F103C8T6 ($1.50 at JLC): no USB DFU in its ROM (USART only), needs a crystal, no CRS; the G0B1 ROM DFU is what removes the second USB port. Swap it in if you have an ST-Link and F103s.
Spare: PA2, PA3, PA15, PB2, PB10-PB15, PC0-PC5, PC8-PC15, PD4-PD9, PF0, PF1 (NC).

B Decoupling



All X7R. One 100 nF per VDD pin, 1 uF + 100 nF at VREF+ (C17, C18).
Place within 2 mm of the pins.

C Reset, BOOT0 (DFU), SWD



BOOT0 shares PA14 (SWCLK) on the STM32G0. Hold BOOT, tap RESET: the ROM bootloader enumerates as USB DFU. Once, with STM32CubeProgrammer, clear option bit nBOOT_SEL so the pin is honoured (the factory default ignores it).

D Debug UART and heartbeat LED

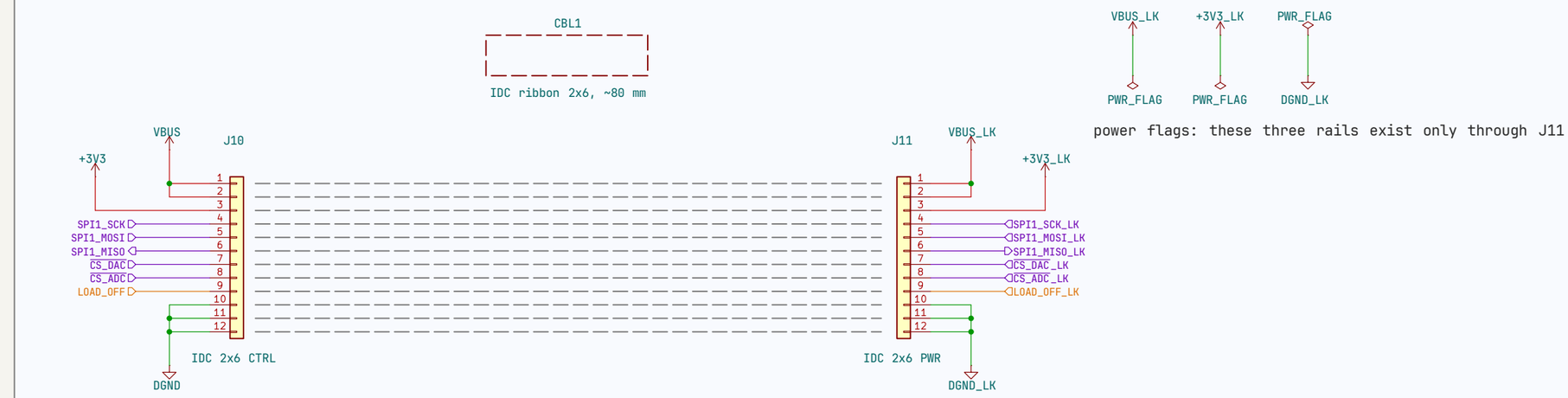


DC Load Box
Sheet: /MCU/
File: mcu.kicad_sch
Title: MCU: STM32G0B1RBT6 (LQFP64, the G0B1 JLC stocks), USB device, SPI
Size: A3 | Date: 2026-11-03 | Rev: E
KiCad E.D.A. 10.0.6 | Id: 4/14

Board link: control board J10 to power board J11 over a 12-way IDC ribbon cable

Two boards on one V-scored fabrication panel. Only USB-side digital and 5 V / 3.3 V power cross the cable; the isolation barrier stays on the power board.

A J10 (control board, back face) - 2x6 IDC ribbon - J11 (power board, USB-side island)



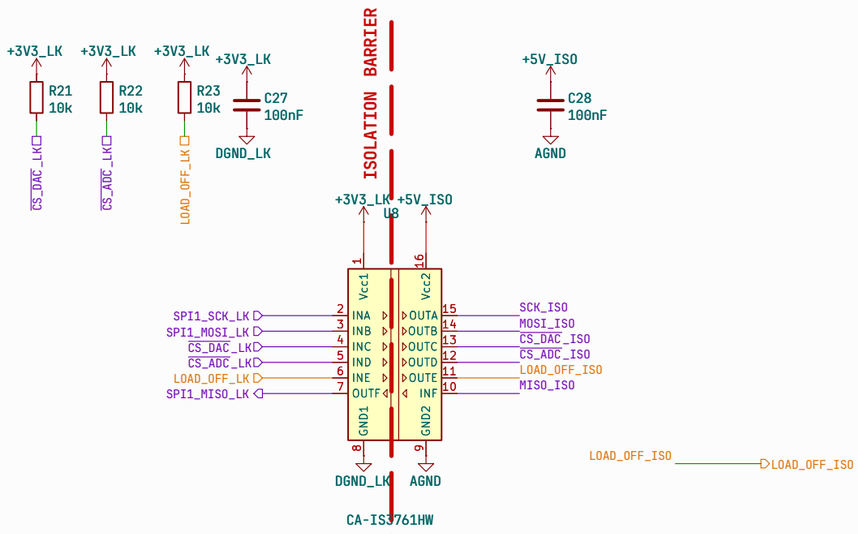
Pins 1 2 VBUS 3 +3V3 4 SCK 5 MOSI 6 MISO 7 /CS_DAC 8 /CS_ADC 9 LOAD_OFF 10 11 12 DGND.
The two boards are separate nets in the PCB (no ratsnest across the V-score): J11-side nets carry the _LK suffix and CBL1 is a BOM line only. The DC-DC and the isolator sit on the power board, so the cable carries nothing isolated and nothing analog; 2 MHz SPI over 80 mm of ribbon needs no termination. Keyed 2x6 box headers (BOOMELE 2.54-2*6P, C9136, both vertical): J10 is hand-soldered on the BACK face of the control board so the ribbon leaves rearward, folds once and plugs into J11 from above on the power board. Check pin 1 to pin 1 with a meter on the first pair; if the key makes that impossible, re-crimp one socket the other way.

DC Load Box	
Sheet: /Board Link/	
File: board_link.kicad_sch	
Title: Board link: control board J10 to power board J11 over a 12-way IDC	
Size: A3	Date: 2026-11-03
KiCad E.D.A. 10.0.6	Id: 5/14

Isolated interface: digital isolator, set-point DAC, readback ADC

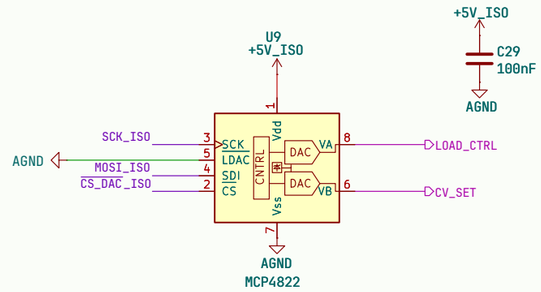
SPI crosses the barrier. The DAC and ADC carry their own references, so no precision reference or trimmer is needed.

A Digital isolator



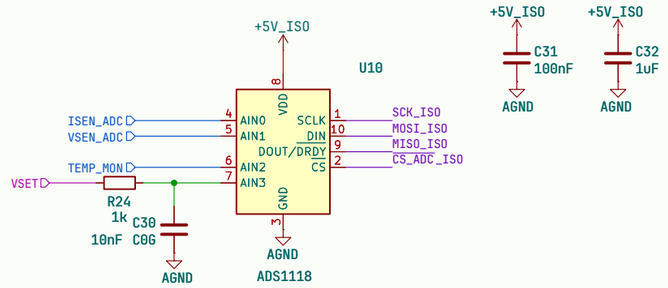
Side 1 = USB side after the J10/J11 cable (3.3 V_LK), side 2 = analog (5 V). Five channels cross, MISO comes back. CA-IS3761HW (the H suffix) and the TI ISO7761 are pin-compatible; both have outputs that default HIGH when side 1 is unpowered or an input floats: chip selects stay de-asserted and LOAD_OFF_ISO is asserted, so the load is OFF until firmware says otherwise. R21-R23 (on this board) give the same guarantee with the MCU in reset or the cable unplugged. 11 ns delay: SPI ⩽ 2 MHz.

B Set-point DAC: A = current ceiling, B = CV voltage



12 bit, internal 2.048 V reference, gain 2: 0 - 4.095 V in 1 mV steps. A: LOAD_CTRL, 2.5 A per volt = 2.44 mA per LSB. B: CV_SET in VSEN volts. Both outputs are 0 V after power-up (POR shutdown): no EEPROM surprise like the MCP4725.

C Readback ADC: 16 bit, 4 channels



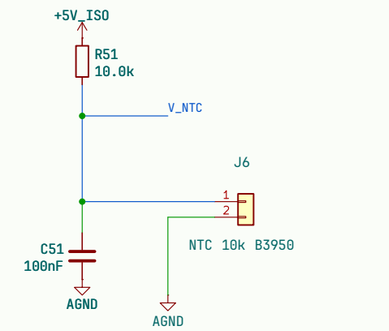
AIN0 ISEN (0.30 V/A), AIN1 VSEN (0.049 V/V), AIN2 heatsink NTC, AIN3 the live V_SET node (PGA +-256 mV): shows what the loops really get after OTP, OFF, KILL or CV pull it down. Internal reference 0.01 %, 7 ppm/K, 860 SPS, plus a die temperature sensor.

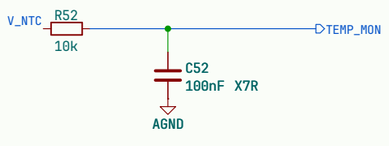
DC Load Box
Sheet: /Isolated Interface/
File: iso_interface.kicad_sch
Title: Isolated interface: digital isolator, set-point DAC, readback ADC
Size: A3 | Date: 2026-11-03 | **Rev: E**
KiCad E.D.A. 10.0.6 | Id: 6/14

Protection: over-temperature, dropout / KILL, NTC-open, over-voltage, hardware OFF

LM339 open-collector comparators on +12V_ISO. Thresholds are ratios of +5V_ISO, the same rail that feeds the NTC divider.

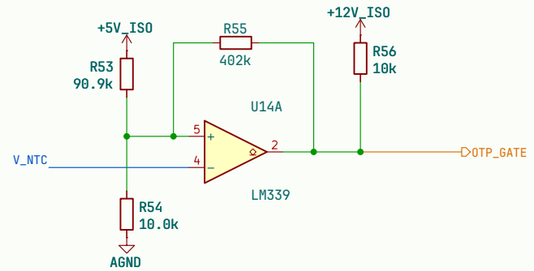
A Heatsink NTC and TEMP_MON





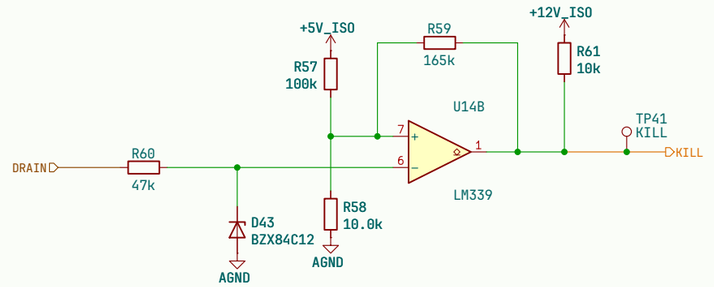
V_NTC (10k B3950 lug thermistor on the heatsink next to FET 2): 2.50 V at 25 C, 1.52 V at 45 C, 0.75 V at 70 C, 0.49 V at 85 C. TEMP_MON goes to the ADC: the MCU runs the fan from it and shows the temperature.

B Over-temperature: 85 C trip, 70 C release



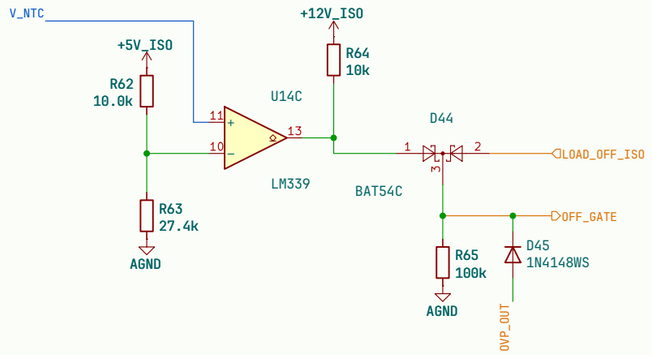
In+ = 0.099 x rail (0.49 V = 85 C). Output high → Q11 shorts V_SET. Hysteresis via R55 from the 12 V output: releases at 0.75 V = 70 C.

C Dropout / anti-wind-up KILL: DRAIN below 0.45 V



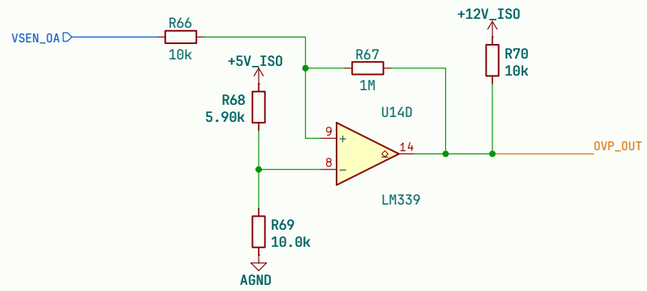
KILL pulls every channel In- high through 4.7k + 1N4148 when the DUT is absent or below 0.45 V; releases above 0.54 V. Stops the 115 - 170 A recovery spikes of deck 04. R60 sees 60 V: 1206.

D NTC open and hardware OFF: all roads to OFF_GATE



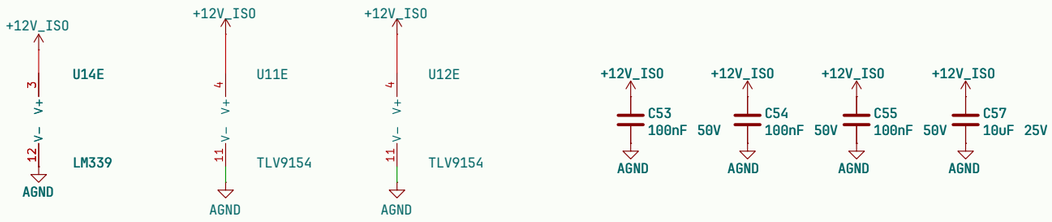
NTC open or unplugged: V_NTC above 0.73 x rail (= 3.66 V, NTC > 27 k) forces 0 A; an open NTC would otherwise read as cold. False trip only below about 3 C. LOAD_OFF_ISO: the MCU-side OFF, 5 V logic through the isolator, asserted whenever the USB side is dead. OVP_OUT (block E) joins through D45. R65 defines the node when nothing drives it.

E Over-voltage: DUT above 62 V



VSEN_OA above 0.629 x rail (3.15 V = 62 V at the terminals) forces 0 A through OFF_GATE. Protects the 58 V fuse rating and the 70 V TVS. R67 gives about 1.8 V of hysteresis. Firmware latches the trip; the comparator alone would re-arm when the DUT sags.

F Supplies of the analog ICs



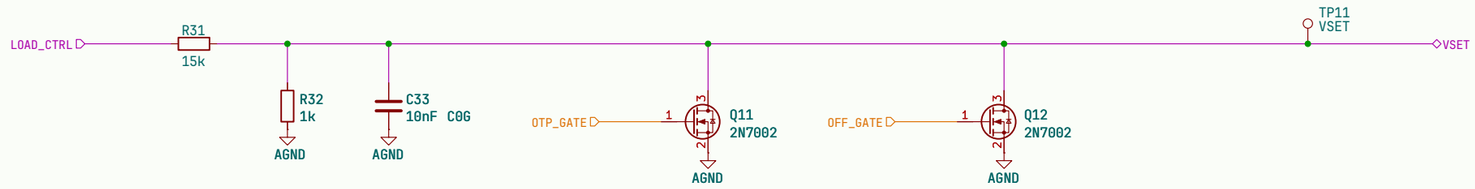
100 nF 50 V at every V+ pin (LM339, U11, U12), 10 uF bulk near U11.

DC Load Box		
Sheet: /Protection/		
File: protection.kicad_sch		
Title: Protection: over-temperature, dropout / KILL, NTC-open, over-voltage,		
Size: A3	Date: 2026-11-03	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 7/14

Set-point and hardware CV: V_SET divider, protective pull-downs, CV error amplifier

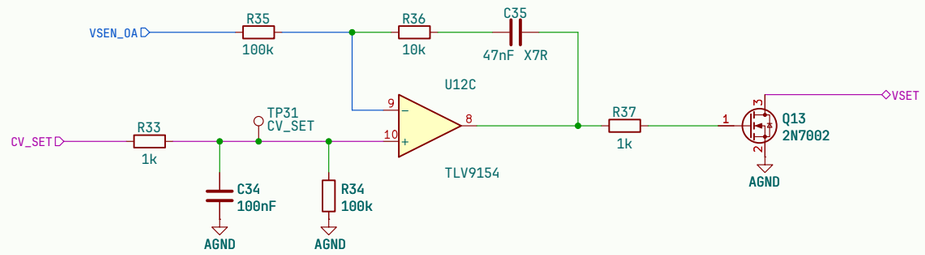
V_SET = LOAD_CTRL / 16 is the only node the four channels follow. Anything that must stop the load pulls it to AGND.

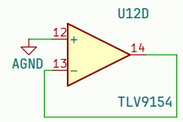
A V_SET: 0 - 256 mV = 0 - 10.24 A



R31 / R32 = 15 : 1, 1 % (the gain is calibrated): 4.096 V in gives 256 mV = 2.56 A per channel. C33 slews the step in 9 us so the loops never see a step edge. Q11 and Q12 are hardware and work with the MCU dead. CV (block B) pulls this node down too, so LOAD_CTRL stays the ceiling in CV.

B Hardware constant-voltage loop (LTspice rev_e deck 06)





spare section: follower at AGND

Why: a current-source DUT cannot be held in CC (two current sources in series rail to compliance), and a firmware CV loop on the DAC cannot either: 1 LSB into the 2.2 uF snubber slews 11 V in one 10 ms update. Here: V_DUT below CV_SET → U12C rises → Q13 pulls V_SET down → less current. CV_SET = 0 V: VSEN_OA (85 mV minimum) > CV_SET, U12C stays low, Q13 off, plain CC (decks 01-04). Deck 06: settles for 0.1 - 8 A with and without 47 uF across the DUT. Below 0.1 A it hunts: use CC or a resistor there.

C Pass-through pins for the block diagram on page 1



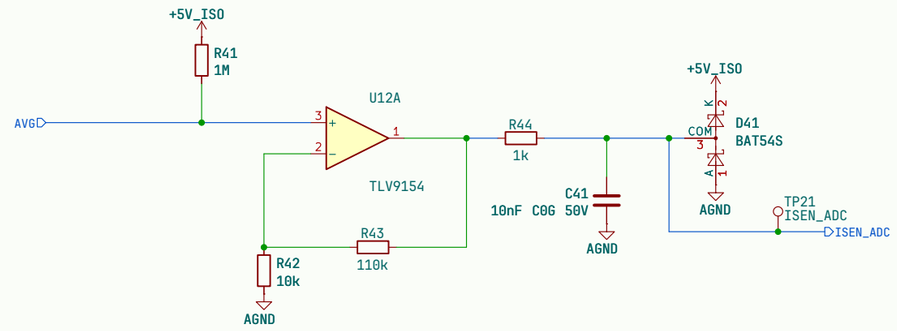
Same nets, second names: VSET_MON feeds the ADC read-back (page 6), VSEN_OA_OVP hands the VSEN amplifier output on to the over-voltage comparator (page 9). They exist so the top sheet can be drawn as a block diagram with wires, without a net label.

DC Load Box
Sheet: /Setpoint & CV/
File: setpoint_cv.kicad_sch
Title: Set-point and hardware CV: V_SET divider, protective pull-downs, CV e
Size: A3 | Date: 2026-11-03 | Rev: E
KiCad E.D.A. 10.0.6 | Id: 8/14

Sense: ISEN = mean of the four source nodes, VSEN = true 4-wire difference amplifier

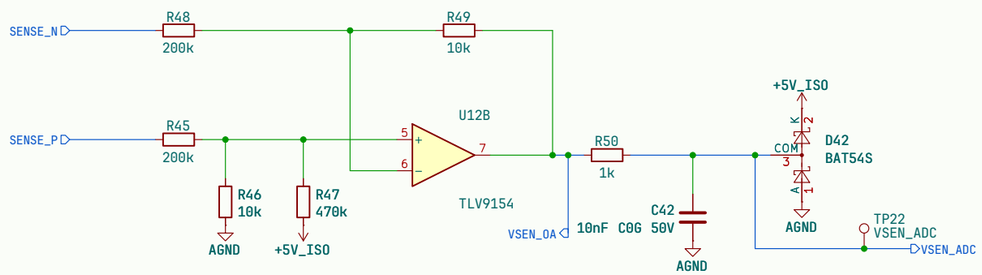
Both outputs are scaled for the 4.096 V ADC range and clamped to +5V_ISO. Offsets are deliberate and calibrated out in firmware.

A ISEN: 0.300 V/A + 0.15 V



Gain 12 (R43 / R42 + 1, 1 %: up to 2 % gain error, removed by the 2-point calibration). R41 injects +12 mV at the input (= +0.15 V out), kept from Rev C: a zero calibration removes it. Full scale 3.22 V at 10.24 A, 0.4 mA per ADS1118 LSB. U12A, standard quad pinout.

B VSEN: 0.049 V/V + 0.10 V, true 4-wire



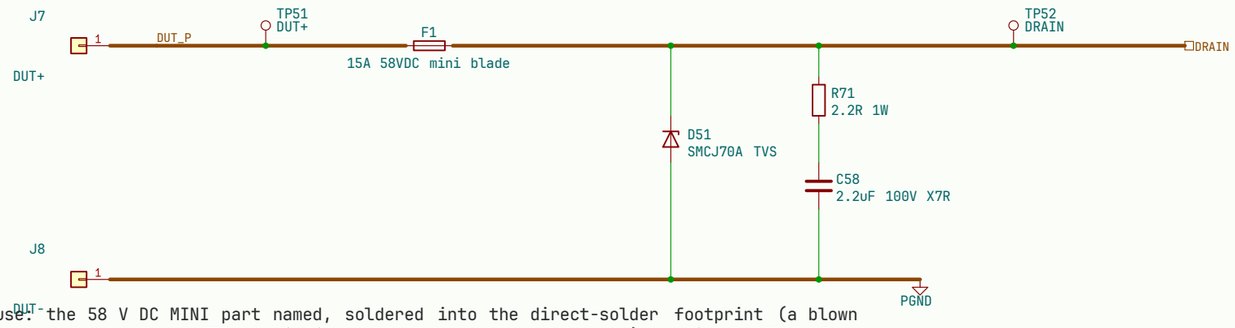
VSEN = 0.049 x (SENSE_P - SENSE_N) + 0.10 V: 2.5 mV of DUT per LSB, clips at 65 V. R45 and R48 see 60 V: 1206. 1 % resistors: up to 2 % gain error (calibrated out) and CMRR about 40 dB, harmless because SENSE_N sits within millivolts of AGND. R49 feeds back to -IN (Rev B had it on +IN and latched). VSEN_OA also feeds CV and OVP.

DC Load Box		
Sheet: /Sense/		
File: sense.kicad_sch		
Title: Sense: ISEN = mean of the four source nodes, VSEN = true 4-wire dif		
Size: A3	Date: 2026-11-03	Rev: E
KiCad E.D.A. 10.0.6		Id: 9/14

Power path: DUT terminals, 58 V fuse, TVS, snubber, 4-wire sense

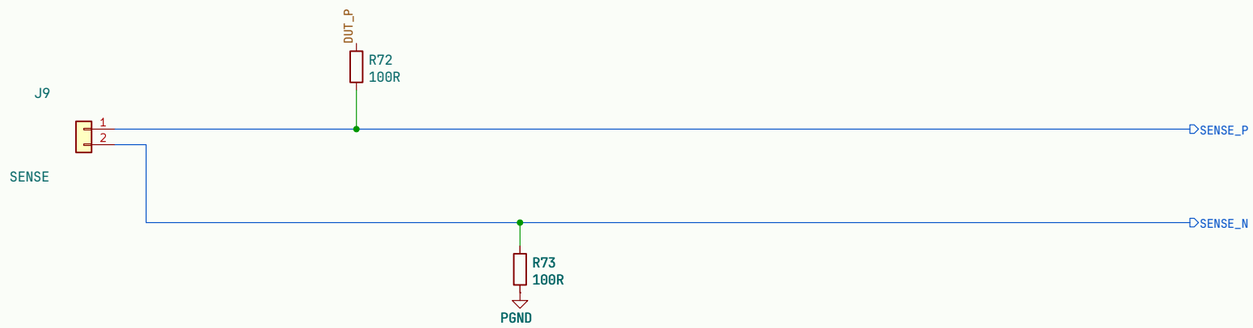
DUT+ → fuse → DRAIN bus → four drains. The source resistors return to PGND, which meets AGND only at NT1.

A Terminals and DRAIN bus



Fuse: the 58 V DC MINI part named, soldered into the direct-solder footprint (a blown fuse means a shorted FET, so it is replaced on the bench anyway). Ordinary blade fuses are rated 32 V, so 58 V is the DUT limit of this board (OVP trips at 62 V). No series diode, so the load works down to 0.6 V. Reverse polarity conducts through the FET body diodes and the TVS and clears the fuse. Copper: ⩾ 5 mm of 2 oz on both layers for DUT+ / DRAIN / PGND. C58 and D51 directly across DRAIN / PGND at the terminals.
J7 / J8: plated 4.3 mm holes with a 10 mm pad, M4 bolt and ring lug; the panel binding posts (if fitted) are wired to the same bolts.

B 4-wire sense with local fallback



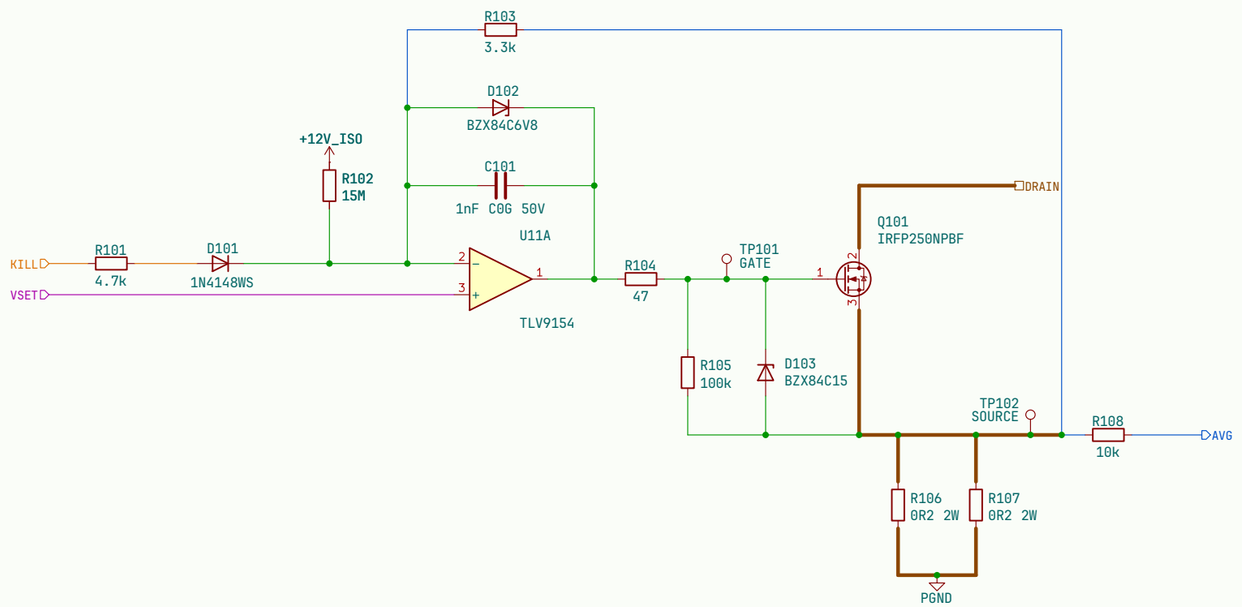
J9 open: R72 / R73 connect the sense amplifier to the terminals. J9 wired to the DUT: they carry only microamps and the lead drop leaves the reading. A miswired sense lead burns a 100 R 1206 open and nothing else.

DC Load Box		
Sheet: /Power Path/		
File: power_path.kicad_sch		
Title: Power path: DUT terminals, 58 V fuse, TVS, snubber, 4–wire sense		
Size: A3	Date: 2026–11–03	Rev: E
KiCad E.D.A. 10.0.6		Id: 10/14

Current-sink channel (4 instances, one TLV9154 section each)

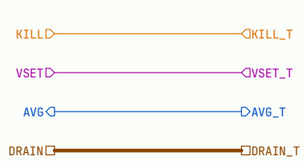
Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): TLV9154 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 (15M) gives +2.6 mV offset, above the TLV9154 0.83 mV Vos limit, so DAC code 0 is truly 0 A (dead zone about 106 mA in total, 42 codes). Loop values from rev_e deck 03: PM ≥ 87 deg, GM ≥ 22 dB for the IRFP250N / 240 / 530 (the big-die IXTH proxy drops to 80 deg / 9 dB). Change them only with a re-run of deck 03.
(n = channel number)

Chain pins: the four channels are daisy-chained on page 1 (each net enters on the left and leaves on the right)



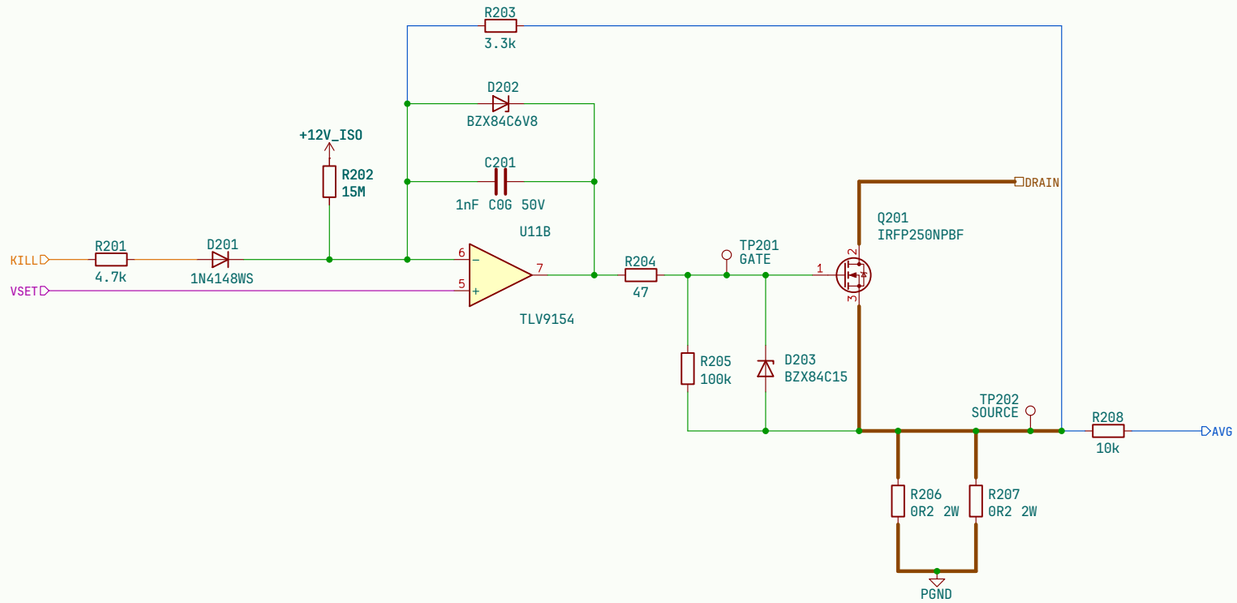
Same nets, second names (KILL_T = KILL ...). Channel 4 leaves KILL_T, VSET_T and AVG_T open; its DRAIN_T carries DRAIN on to the dropout comparator. Electrically every channel still hangs directly on the four nets.

DC Load Box		
Sheet: /Channel 1/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one TLV9154 section each)		
Size: A3	Date: 2026-11-03	Rev: E
KiCad E.D.A. 10.0.6		Id: 11/14

Current-sink channel (4 instances, one TLV9154 section each)

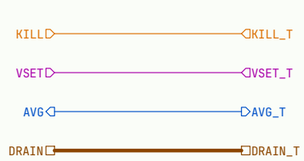
Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): TLV9154 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 (15M) gives +2.6 mV offset, above the TLV9154 0.83 mV Vos limit, so DAC code 0 is truly 0 A (dead zone about 106 mA in total, 42 codes). Loop values from rev_e deck 03: PM ≥ 87 deg, GM ≥ 22 dB for the IRFP250N / 240 / 530 (the big-die IXTH proxy drops to 80 deg / 9 dB). Change them only with a re-run of deck 03.
(n = channel number)

Chain pins: the four channels are daisy-chained on page 1 (each net enters on the left and leaves on the right)



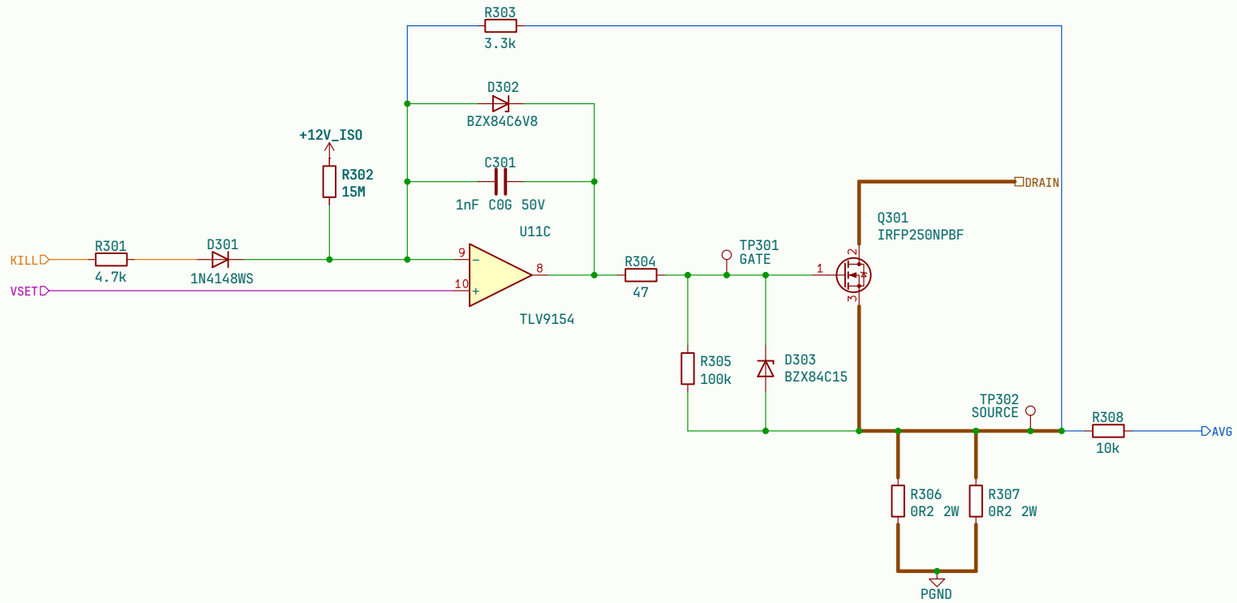
Same nets, second names (KILL_T = KILL ...). Channel 4 leaves KILL_T, VSET_T and AVG_T open; its DRAIN_T carries DRAIN on to the dropout comparator. Electrically every channel still hangs directly on the four nets.

DC Load Box		
Sheet: /Channel 2/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one TLV9154 section each)		
Size: A3	Date: 2026-11-03	Rev: E
KiCad E.D.A. 10.0.6		Id: 12/14

Current-sink channel (4 instances, one TLV9154 section each)

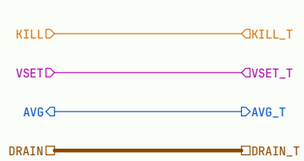
Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): TLV9154 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 (15M) gives +2.6 mV offset, above the TLV9154 0.83 mV Vos limit, so DAC code 0 is truly 0 A (dead zone about 106 mA in total, 42 codes). Loop values from rev_e deck 03: PM ≥ 87 deg, GM ≥ 22 dB for the IRFP250N / 240 / 530 (the big-die IXTH proxy drops to 80 deg / 9 dB). Change them only with a re-run of deck 03.
(n = channel number)

Chain pins: the four channels are daisy-chained on page 1 (each net enters on the left and leaves on the right)



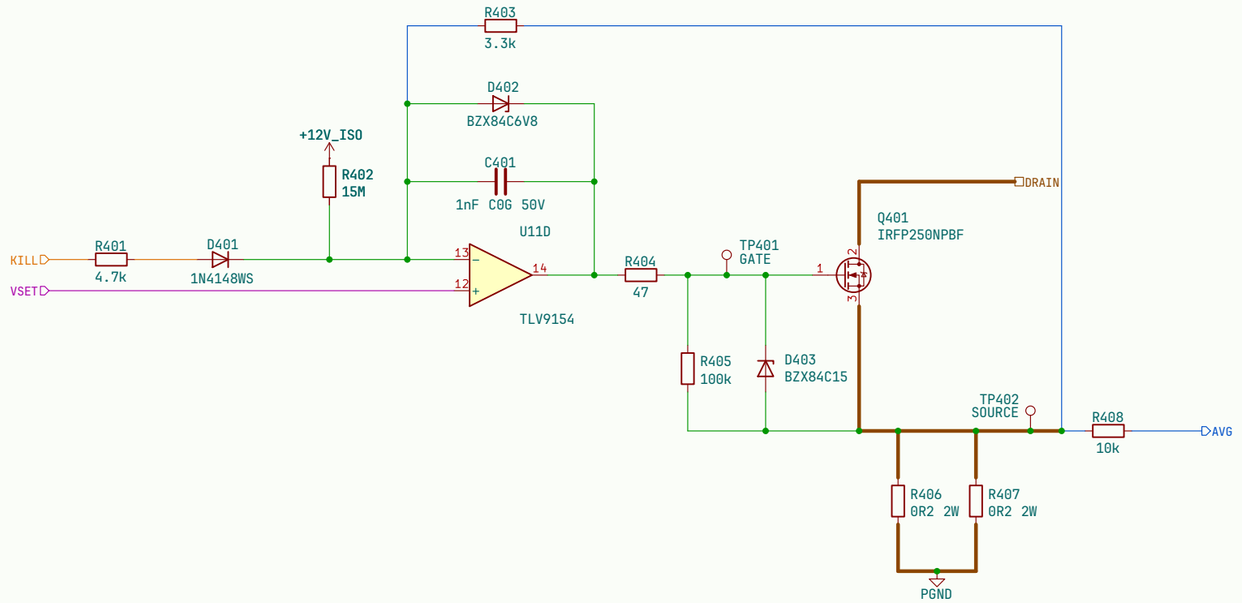
Same nets, second names (KILL_T = KILL ...). Channel 4 leaves KILL_T, VSET_T and AVG_T open; its DRAIN_T carries DRAIN on to the dropout comparator. Electrically every channel still hangs directly on the four nets.

DC Load Box		
Sheet: /Channel 3/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one TLV9154 section each)		
Size: A3	Date: 2026-11-03	Rev: E
KiCad E.D.A. 10.0.6		Id: 13/14

Current-sink channel (4 instances, one TLV9154 section each)

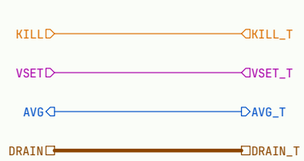
Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): TLV9154 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 (15M) gives +2.6 mV offset, above the TLV9154 0.83 mV Vos limit, so DAC code 0 is truly 0 A (dead zone about 106 mA in total, 42 codes). Loop values from rev_e deck 03: PM ≥ 87 deg, GM ≥ 22 dB for the IRFP250N / 240 / 530 (the big-die IXTH proxy drops to 80 deg / 9 dB). Change them only with a re-run of deck 03.
(n = channel number)

Chain pins: the four channels are daisy-chained on page 1 (each net enters on the left and leaves on the right)



Same nets, second names (KILL_T = KILL ...). Channel 4 leaves KILL_T, VSET_T and AVG_T open; its DRAIN_T carries DRAIN on to the dropout comparator. Electrically every channel still hangs directly on the four nets.

DC Load Box		
Sheet: /Channel 4/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one TLV9154 section each)		
Size: A3	Date: 2026-11-03	Rev: E
KiCad E.D.A. 10.0.6		Id: 14/14